

## Assignment - 02

DATE: 28/10/23

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(Computer Network)

(22BCON1068)

Section - A (2x5=10)

Q1. What does CDMA stand for in the context of the link layer's medium access control.

Ans. CDMA stands for code division multiple access in the context of the link layer's medium access control (MAC). It is a channel access method that allows multiple users to transmit data simultaneously over the same frequency channel. This is achieved by using unique code sequences to spread out the data of each user across the entire channel bandwidth.

CDMA is a spread spectrum technique, which means that it spreads the signal over a wider bandwidth that is necessary to transmit the information. This makes the signal more resistant to interference and noise.

In the context of link layer's MAC, CDMA provides a way for multiple users to access the same medium without colliding with each other. This is important for wireless networks, where the medium is shared by all users.

→ Here is an example of how CDMA works in the link-layer's MAC.

- (i) Each user is assigned a unique code sequence.
- (ii) When a user wants to transmit data, it spreads its data across the entire channel bandwidth using its code sequence.
- (iii) The receiver despreads the received signal using the code sequences of all users.

Q2. Name one disadvantage of the ALOHA multiple access protocol.



Ans: One disadvantage of the ALOHA multiple access protocol is that it is prone to collisions. This is because any station can transmit data at any time, and if two or more stations transmit at the same time, their signals will collide and the data will be corrupted.

Q3. What is the primary purpose of link layer addressing?

Ans: The primary purpose of link layer addressing is to uniquely identify devices on a local network segment. This is necessary for devices to communicate with each other on the same segment.

Link layer addresses are typically assigned to devices by the manufacturer. However, some devices can also be manually configured with a link layer address.

Link layer addresses are different from IP addresses. IP addresses are used to identify devices on a network, regardless of which network segment they are on. Link layer addresses are only used to identify devices on the same network segment.

Some examples of link layer addresses:

- Ethernet addresses (MAC addresses)
- Token Ring addresses
- FDDI addresses
- Wireless LAN addresses

Q4. What is the maximum length of a standard Ethernet frame?

Ans: The maximum length of a standard Ethernet frame is 1518 bytes. This includes the Ethernet header (14 bytes), the payload (up to 1500 bytes), and the frame check sequence (FCS) field (4 bytes).

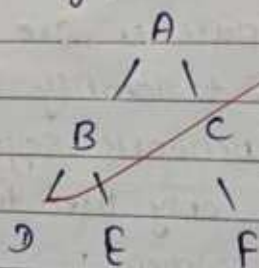


Q5. Define "Spanning Tree" in the context of network topology?

Ans. A spanning tree in the context of network topology is a subset of the network's links that connects all of the network's nodes without creating any loops. A loop is a situation where two or more links connect the same two nodes, which can cause traffic to be broadcast endlessly around the loop.

Spanning trees are created using a variety of algorithms, such as the Spanning Tree Protocol (STP). STP is a distributed protocol that runs on all of the bridges and switches in a network.

Here is an example of a spanning tree in a network topology.



In this example, the spanning tree would consist of the following links:

AB, BC, CD, CE, EF.

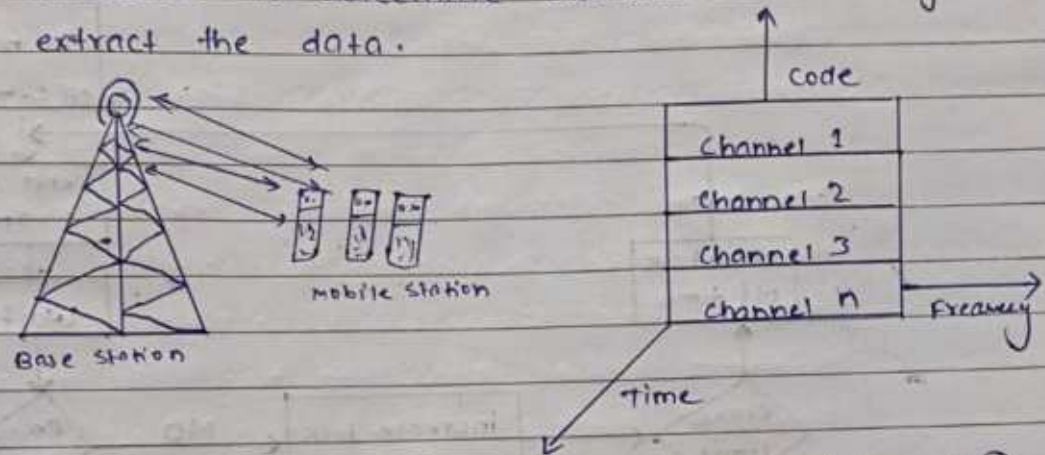
Section-B (3x7=21)

Q1. Compare and contrast CDMA and ALOHA as multiple access protocols. Highlight their key differences.

Ans. CDMA (Code Division Multiple Access) and ALOHA are both multiple access protocols that allow multiple devices to share a single communication channel. However, they do so in different ways.



- CDMA uses spread spectrum technology to spread the signal of each device across a wide frequency band. This allows multiple devices to transmit simultaneously without interfering with each other. Each device is assigned a unique code sequence, which is used to modulate its signal. The receiver uses the same code sequence to determine demodulate the signal and extract the data.



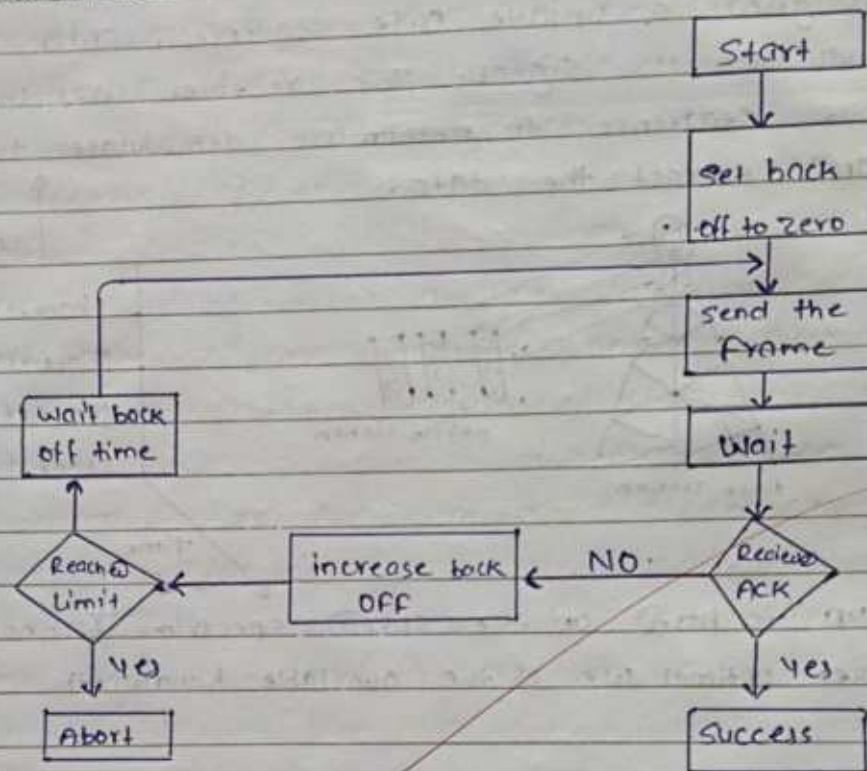
CDMA is based on the spread spectrum technology and makes optimal use of the available bandwidth.

- ALOHA :- ALOHA is a simpler Protocol that does not use spread spectrum technology. Instead, it relies on devices to transmit randomly and to retransmit if their transmission collides with another device's transmission.

Aloha is a multiple access Protocol for transmission of data via a shared network channel. It operates in the medium access sublayer (MAC sublayer) of the Open systems interconnection (OSI) model. Using this Protocol, several data streams originating from multiple nodes are transferred through a multi-point transmission channel.

in ALOHA, each node or station

transmits a frame without trying to detect whether the transmission channel is idle or busy. If the channel is idle, then the frames will be successfully transmitted.



(Protocol flow chart for ALOHA)

#### • Advantages of CDMA :-

- More efficient than ALOHA
- Can support more user
- More resistant to interference

#### • Disadvantages of CDMA :-

- More complex than ALOHA
- Requires ~~less~~ More sophisticated equipment



### • Advantages of ALOHA:-

- Simple to implement
- Requires less sophisticated equipment

### • Disadvantages of ALOHA:-

- Less efficient than CDMA
- Can support fewer users
- More susceptible to interference

### • Application of CDMA:-

- Wireless networks LANs
- Cellular networks
- Satellite communications
- GPS

### • Application of ALOHA:-

- Wireless network
- Satellite comm.
- RFID systems

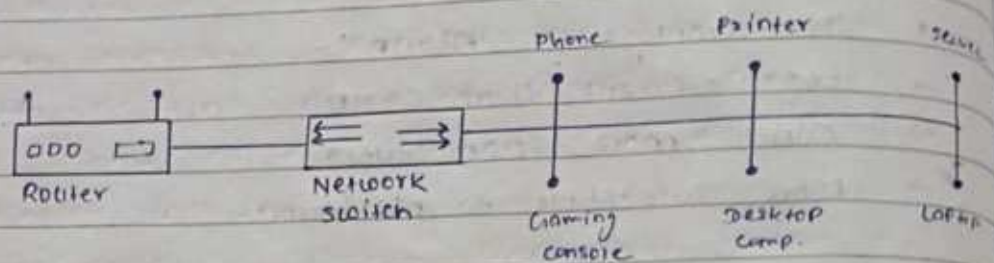
Q2. Explain the concept of Link layer switching and how it improves network performance.

Ans. Link layer switching is a network switching technology that works at the data link layer (layer 2) of the OSI model. It forwards data frames based on the MAC addresses of the sources and destination devices.

When a link layer switch receives a data frame, it examines the MAC address of the destination device. If the switch knows the location of the destination device, it forwards the frame to that



port. If the switch does not know the location of the destination device, it floods the frame to all ports on the switch.



- Link layer switching Improves network performance in several ways :-
  - Reduces Broadcast traffic :- When a device sends a broadcast frame, it is sent to all devices on the network. This can cause significant congestion on the network, especially if there are a large number of devices. Link layer switches can reduce broadcast traffic by only forwarding broadcast frames to the ports that need to receive them.
  - Reduces Latency :- Latency is the time it takes for a data frame to travel from the source device to the destination device. Link layer switches can reduce latency by forwarding data frames directly to the destination device, without having to send them to a router.
  - Improves Scalability :- Link layer switches can help to improve the scalability of a network by allowing more devices to be connected to the network without sacrificing performance.
- Link layer switching is a widely used technology in both wired and wireless networks.



It is a key component of high-performance networks such as those used in enterprise and data center environments.

Q3. Describe the Channel Allocation Problem in wireless communication. What are the main challenges associated with channel allocation in a wireless network?

Ans. The channel allocation problem in wireless communication is the problem of assigning radio frequency channels to users in a way that maximizes throughput and minimizes interference.

The main challenges associated with channel allocation in a wireless network are :-

- Limited spectrum resources : The radio frequency spectrum is a limited resource, and there are many different users who need to access it. This means that channel allocation must be done efficiently to ensure that all users can access the spectrum they need.
- Interference : Radio waves can interfere with each other, so channel allocation must be done in a way that minimize interference between users.
- Dynamic channel conditions : Wireless channels are dynamic and can change over time due to factors such as fading, shadowing and interference. This means that channel allocation must be adapted to changes in the channel conditions.
- Fairness : Channel allocation should be fair to all users, so that no user is consistently disadvantaged.



channel allocation is a complex problem, and there is no single solution that works for all cases. The best approach to channel allocation will depend on the specific characteristics of the wireless network, such as the type of users, the traffic patterns, and the channel conditions.

Some of the common channel allocation strategies include:

- Fixed channel allocation: Each user is assigned a specific channel, which they use all the time. This is a simple approach, but it is not very efficient, as it can lead to channel underutilization.
- Dynamic channel allocation: Channels are assigned to users on a demand basis. This is a more efficient approach, but it is also more complex.
- Hybrid channel allocation: A combination of fixed and dynamic channel allocation is used. This approach can provide a good balance between efficiency and simplicity.

Channel allocation is an important part of wireless network design and operation. By carefully allocating channels to users, network operators can improve throughput, reduce interference and ensure that all users have fair access to the spectrum.

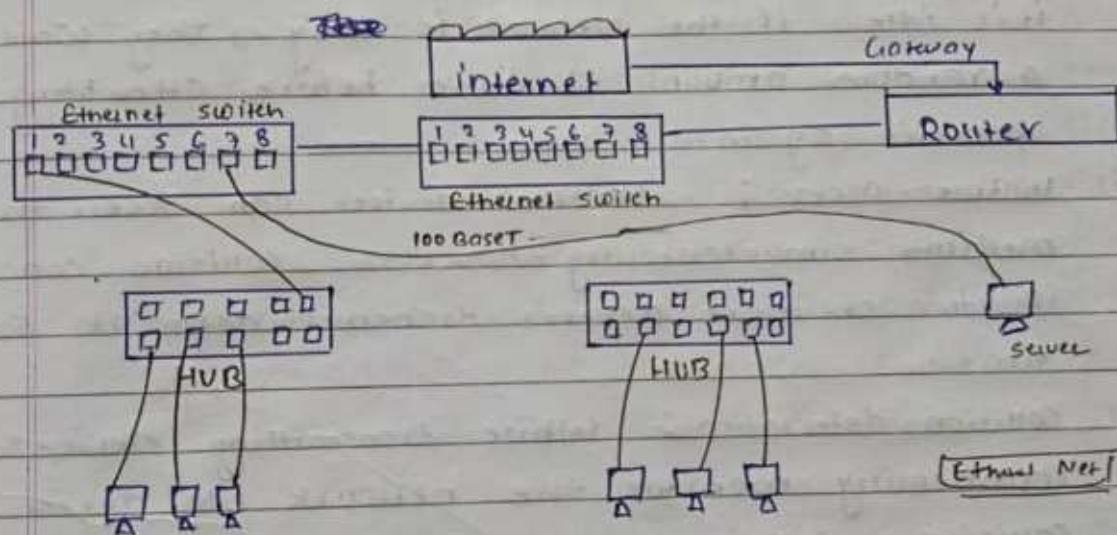
- PART section - c ( $3 \times 11 = 33$ )

Q1. Explain the principles of Ethernet as a multiple access protocol. Discuss its collision detection mechanism and how it handles collisions.



Ans. Ethernet is a widely used multiple access protocol for local area networks (LANs). Multiple access protocols are designed to allow multiple devices to share a common communication medium, such as a network cable, without causing conflicts or collisions.

Ethernet is a multiple access protocol, which means that multiple devices can share a single physical medium to transmit data. This is achieved using a protocol called Carrier Sense Multiple Access with Collision Detection (CSMA/CD).



The CSMA/CD Protocol is effective at reducing collision but it is not perfect. There is still a chance of collision occurring if two devices begin to transmit at the same time. This is especially true on networks with a lot of devices or long cable segments.

- Principles of Ethernet as a multiple Access protocol :-
- (i) Contention based :- Devices compete for access to the medium.
- (ii) Distributed :- There is no central authority that controls the medium.



- (iii) Fair :- All devices have an equal chance of accessing the medium.
- (iv) Robust :- The Protocol can handle collisions and recovers from them.
- (v) Scalable :- The Protocol can support a large number of devices on the network.

Ethernet's collision detection mechanism works →

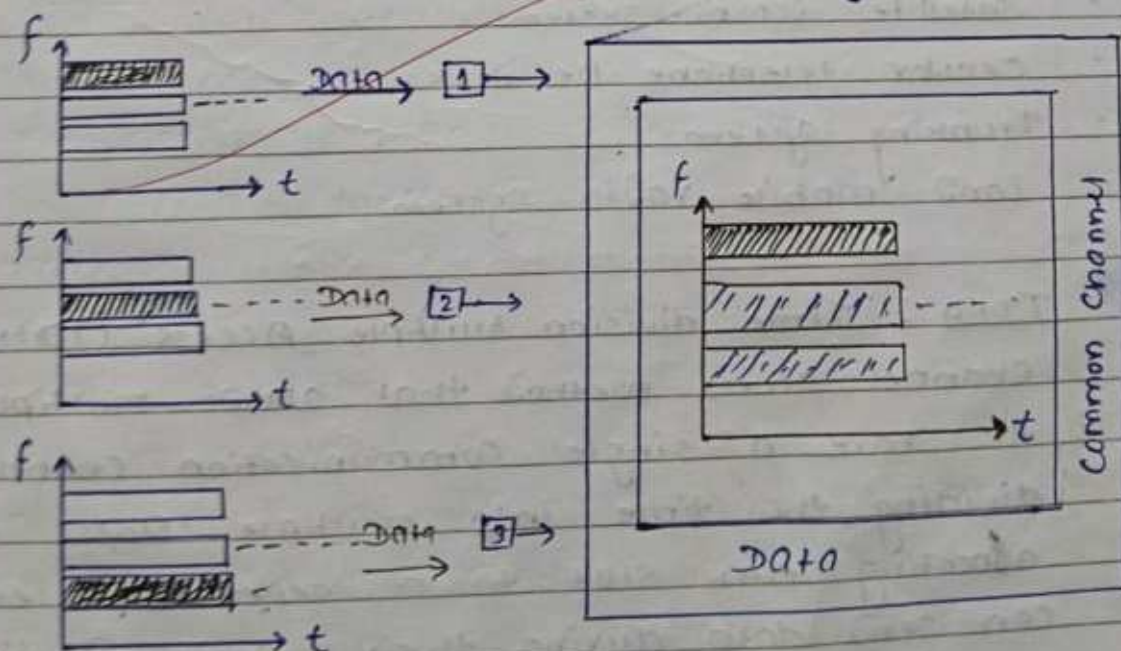
- (i) Carrier Sense :- Before transmitting data, devices using Ethernet listen to the network to check if it is idle. If the network is busy, they wait for a random amount of time before attempting to transmit again.
- (ii) Multiple Access :- Multiple devices can access the Net. medium simultaneously. However, collision can occur if two or more devices transmit data at the same time.
- (iii) Collision Detection :- While transmitting devices continuously monitor the network to detect collision. Collision happens when multiple devices transmit data and their signals overlap.
- (iv) Exponential Backoff :- If multiple collision occurs consecutively, backoff time  $T$  sec exponentially to avoid repeated collision and give more time for network to clear.
- (v) Backoff Algorithms :- After the signal jam, colliding devices wait for a random amount of time before reattempting transmission. This helps to reduce chances of another occurring collisions.



It handles collision by immediately stop transmitting and send a jam signal to notify all other devices on the network about collision. After that each colliding device wait for a random amount of time before reattempting transmission using a backoff algorithm. This helps to reduce the chance of another collision and ensures fair access to the network medium.

Q2. Describe FDMA, TDMA and CDMA with example.

Ans Frequency Division Multiple Access (FDMA) is a channel access method that allows multiple users to share a single communication channel by dividing the bandwidth of the channel into separate non-overlapping frequency sub-channels and allocating each sub-channel to a separate user. Users can send data through a subchannel by modulating it on a carrier wave at the subchannel's frequency.





- Advantage of FDMA:-
  - simple and efficient
  - low latency
  - can support a large number of users
  - Easy to implement.

- Disadvantages of FDMA:-
  - Requires a large amount of spectrum.
  - Not as efficient as TDMA or CDMA.

- Example of FDMA:-

One example of FDMA is AM radio broadcasting. In AM broadcasting, the radio spectrum is divided into multiple channels, each of which is assigned to a different radio station. Each radio station transmits its signal on a unique channel, and listeners can tune their radios to the frequency of the station they want to listen to.

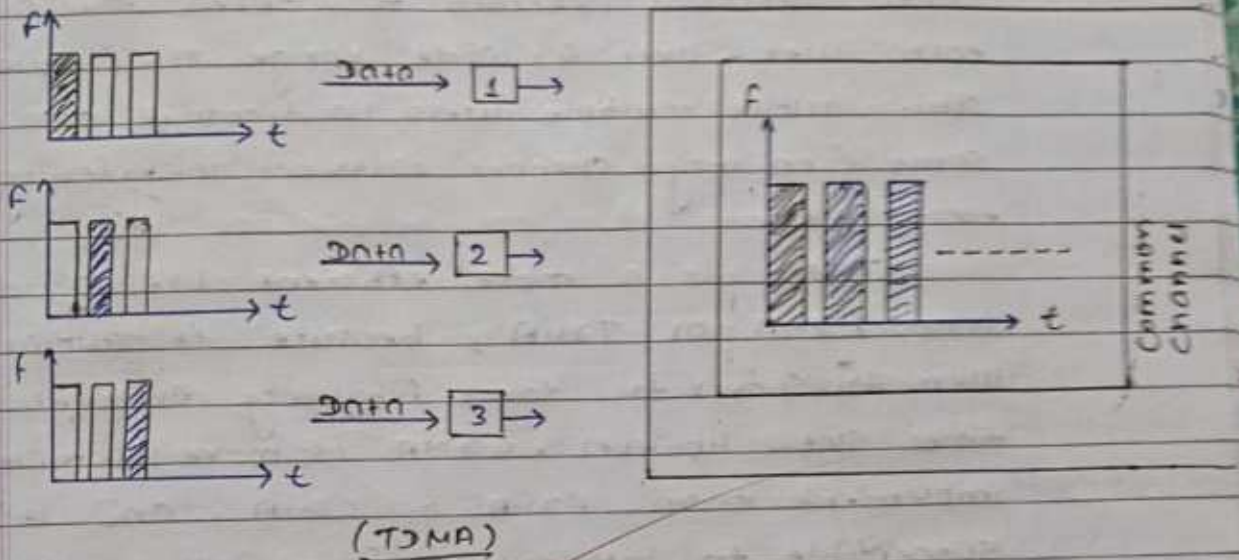
- Application of FDMA:-

- AM and FM radio broadcasting
- Satellite comm. system.
- Cellular telephone networks.
- Trunking systems
- Land-mobile radio systems.

- TDMA : Time division Multiple Access (TDMA) is a channel access method that allows multiple users to share a single communication channel by dividing the time into multiple slots and allocating each slot to a separate user. Users can send data during their assigned time slot, and the channel is idle during the other time slots.



TDMA is a more efficient use of spectrum than FDMA, because it allows multiple users to share the same frequency channel. However, TDMA is more complex to implement than FDMA, and it is more susceptible to latency.



#### • Advantages of TDMA :-

- More efficient use of spectrum than FDMA.
- can support more users on the same spectrum.
- More secure than FDMA.

#### • Disadvantages of TDMA :-

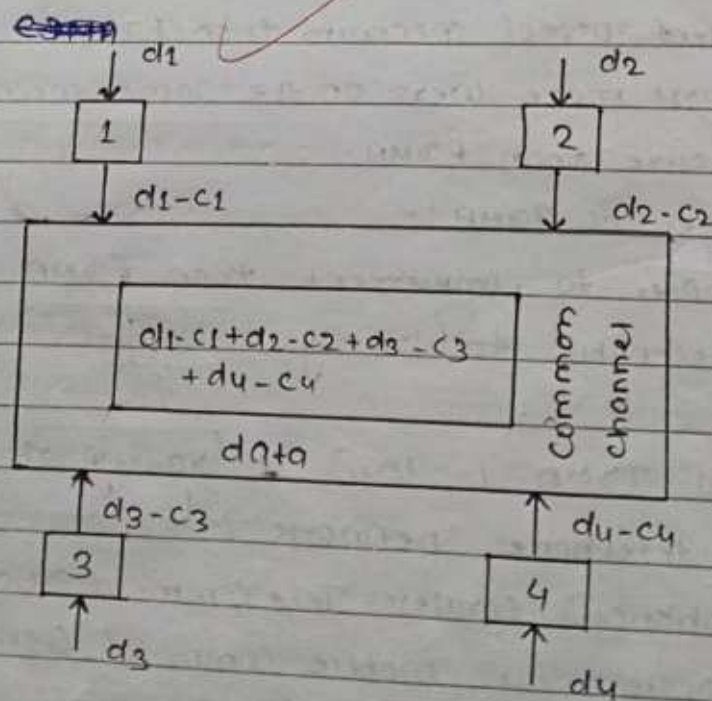
- More complex to implement than FDMA.
- More susceptible to latency

- Example of TDMA is used in variety of applications - like
- Cellular telephone network
- Digital Enhanced Cordless Telecomm. (DECT)
- Global system for mobile comm. (GSM)
- Digital Mobile Radio (DMR)
- IEEE 802.11a/b/g wireless networking.



- CDMA: Code Division Multiple Access (CDMA) is a channel access method that allows multiple users to share a single communication channel by using unique codes to distinguish the signals from different users. CDMA uses a technique called spread spectrum to spread the signal of each user over a wide range of frequencies. This allows multiple users to transmit data on the same frequency channel without interfering with each other.

CDMA is a more efficient use of spectrum than FDMA or TDMA, because it allows multiple users to share the same frequency channel and time slot. However, CDMA is more complex to implement than FDMA or TDMA, and it is more susceptible to interference from other CDMA users.



(CDMA)



- Advantage of CDMA :-

- More efficient use of spectrum
- Can support the most users on the same spectrum
- More secure than FDMA or TDMA

- Disadvantages of CDMA :-

- More complex to implement than FDMA or TDMA
- Can have higher latency than FDMA or TDMA

- Examples of CDMA is used in variety of Application like-

- Cellular telephone networks (3G and 4G)
- GPS
- Bluetooth
- WiMAX

- Difference between FDMA, TDMA and CDMA.

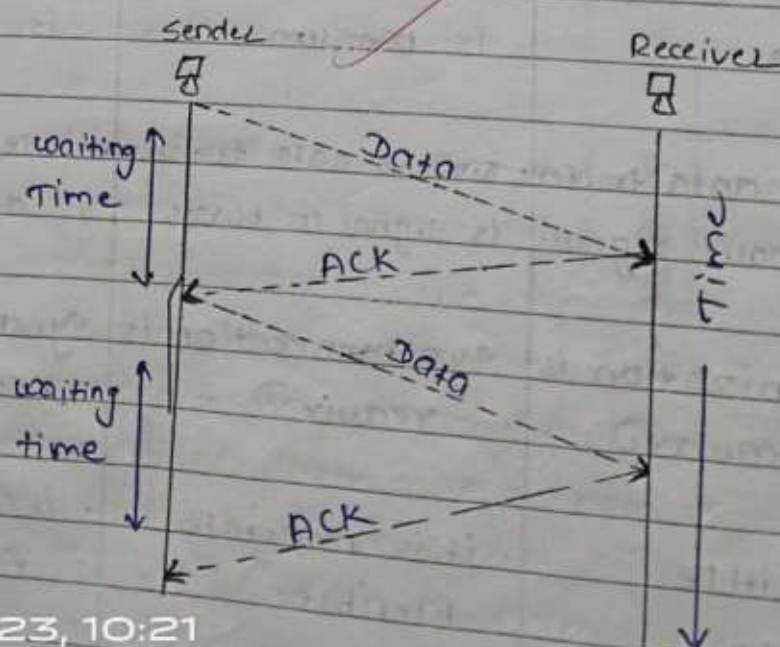
| FDMA                                                                                                    | TDMA                                                                                               | CDMA                                                                                               |
|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• FDMA stands for Frequency Division Multiple Access.</li> </ul> | <ul style="list-style-type: none"> <li>• TDMA stands for Time division multiple Access.</li> </ul> | <ul style="list-style-type: none"> <li>• CDMA stands for code division multiple Access.</li> </ul> |
| <ul style="list-style-type: none"> <li>• There is no need of any Codeword.</li> </ul>                   | <ul style="list-style-type: none"> <li>• There is no need of any Codeword.</li> </ul>              | <ul style="list-style-type: none"> <li>• Codeword is necessary.</li> </ul>                         |
| <ul style="list-style-type: none"> <li>• The rate of data is low.</li> </ul>                            | <ul style="list-style-type: none"> <li>• The rate of data is medium.</li> </ul>                    | <ul style="list-style-type: none"> <li>• The rate of data is high.</li> </ul>                      |
| <ul style="list-style-type: none"> <li>• Mode of data transfer is continuous signal.</li> </ul>         | <ul style="list-style-type: none"> <li>• Mode of data transfer is signal in bursts.</li> </ul>     | <ul style="list-style-type: none"> <li>• Mode of data transfer is digital signal.</li> </ul>       |
| <ul style="list-style-type: none"> <li>• Synchronization is not required.</li> </ul>                    | <ul style="list-style-type: none"> <li>• Synchronization is required.</li> </ul>                   | <ul style="list-style-type: none"> <li>• Synchronization is not required.</li> </ul>               |
| <ul style="list-style-type: none"> <li>• it is little flexible.</li> </ul>                              | <ul style="list-style-type: none"> <li>• it is moderate flexible.</li> </ul>                       | <ul style="list-style-type: none"> <li>• it is highly flexible.</li> </ul>                         |



Q3. write a short notes on —

(a) Stop and wait Protocol :- Stop and wait Protocol is a simple automatic Repeat request (ARQ) Protocol used for reliable data transmission over an unreliable channel. It is a simpler protocol, meaning the data can only be transmitted in one direction at a time. In stop and wait, the sender transmits a single frame and then waits for an acknowledgement (ACK) from the receiver. If the sender does not receive an ACK within a certain timeout period, it re-transmits the frame. Once the receiver receives the frame correctly it sends an ACK back to the sender. The sender then transmits the next frame in the sequence.

Stop and wait is a single protocol, but it is also inefficient. This is because the sender can only transmit one frame at a time, and it has to wait for an ACK before transmitting the next frame.





- Advantages of Stop and Wait Protocol:
  - simple and easy to implement
  - Reliable
  - Efficient for short messages
- Disadvantages of Stop and Wait Protocol:
  - inefficient for long message
  - can lead to high latency

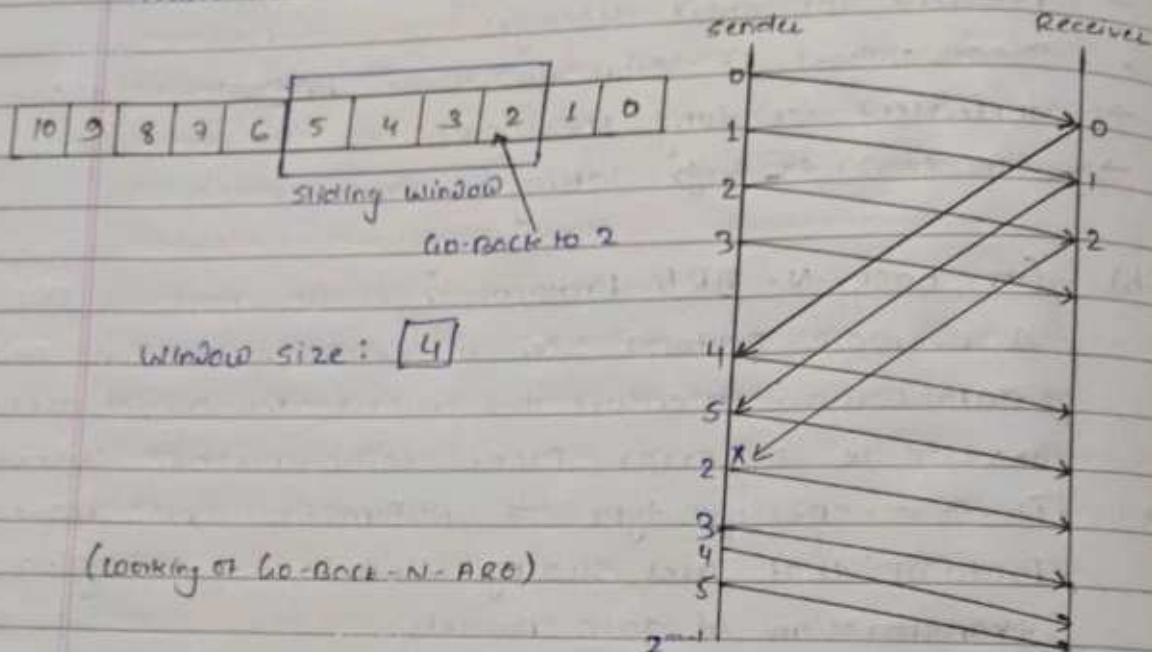
(b) Go-back-N-ARQ Protocol :- Go-back-N-ARQ is a network protocol used in telecommunications and data comm. to ensure the reliable delivery of data over unreliable or error-prone communication channels. it is a specific type of Automatic repeat request (ARQ) Protocol that uses sliding windows for managing the transmission of data packets.

Go-back-N-ARQ Protocol is a reliable data transmission protocol that improves upon the stop and wait protocol by allowing the sender to transmit multiple packets of data before waiting for an acknowledgement from the receiver. this makes it more efficient than the stop and wait protocol, which can lead to high latency for long messages.

- How it works:
- The sender maintains a window of  $N$  packets that it can transmit without waiting for an acknowledgement.
- The sender transmits each packet in the window.
- if the receiver receives a packet correctly, it sends an ACK for that packet.
- If the receiver does not receive a packet correctly, it does not send an ACK.



- If the sender does not receive an ACK for a packet within a certain timeout period, it retransmits all of the packets in the window.



- Advantages of Go-Back-N-ARQ Protocol :-
  - The sender can send many frames at a time
  - Efficiency is more.
- Disadvantages :-
  - Buffer requirement
  - Transmitter needs to store the last N packet.

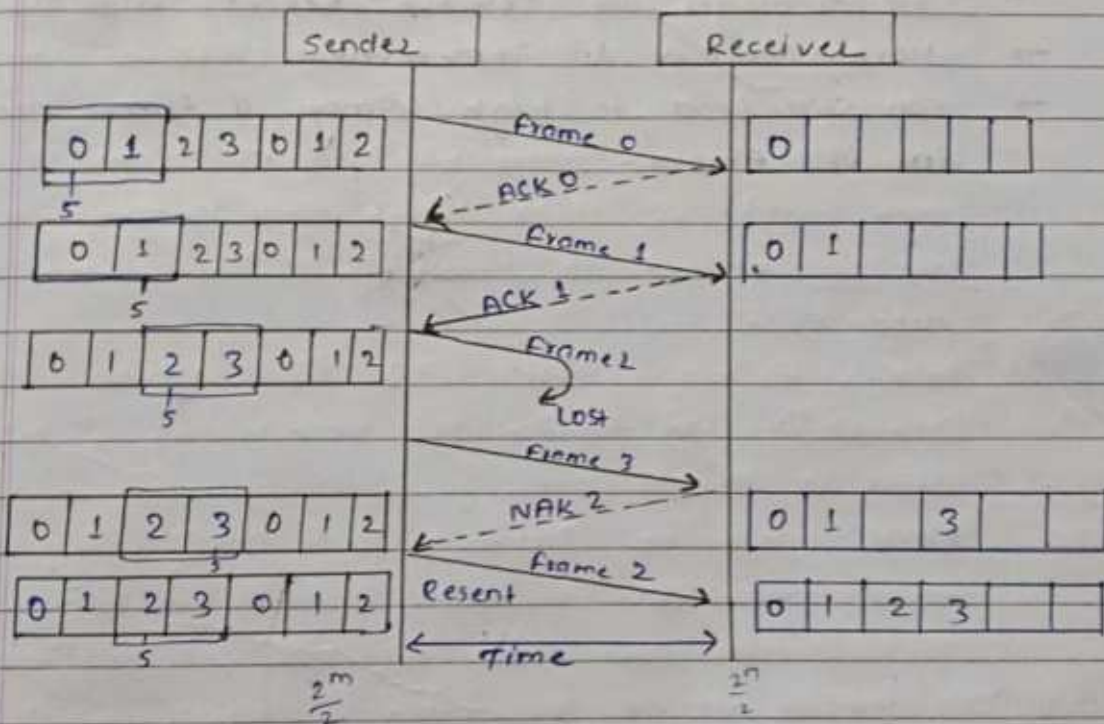
(C) Selective Repeat : Selective Repeat is a reliable data transmission protocol that allows the sender to retransmit only the packets that were not received correctly by the receiver. This makes it more efficient than Go-Back-N-ARQ protocol, especially if there are only a few errors in the transmission.

It is a data link layer protocol uses a sliding window method. The Go-Back



-N-ARQ Protocol works well if it has fewer errors. But if there is a lot of error in the frame, lots of bandwidth loss in sending the frames again. So, we use the selective repeat ARQ Protocol. In this Protocol, the size of sender window is always equal to the size of the receiver window. The size of the sliding window is always greater than 1.

#### • Working of Selective Repeat Protocol:-



#### How it works:-

- The sender maintains a window of  $N$  packets that it can transmit without waiting for an acknowledgment.
- The sender transmits each packet in the window.
- If the receiver receives a packet correctly, it sends an ACK for that packet.
- If the receiver does not receive a packet correctly,

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It sends a NACK for that packet.  
(v) if the sender receives a NACK for a packet it retransmits only that packet.

• Advantages of Selective Repeat Protocol :-

- More efficient, especially if there are only a few errors in the transmission.
- Lower latency.

• Disadvantages of Selective Repeat Protocol :-

- More complex to implement.
- Can still lead to high latency if there are a lot of errors.

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